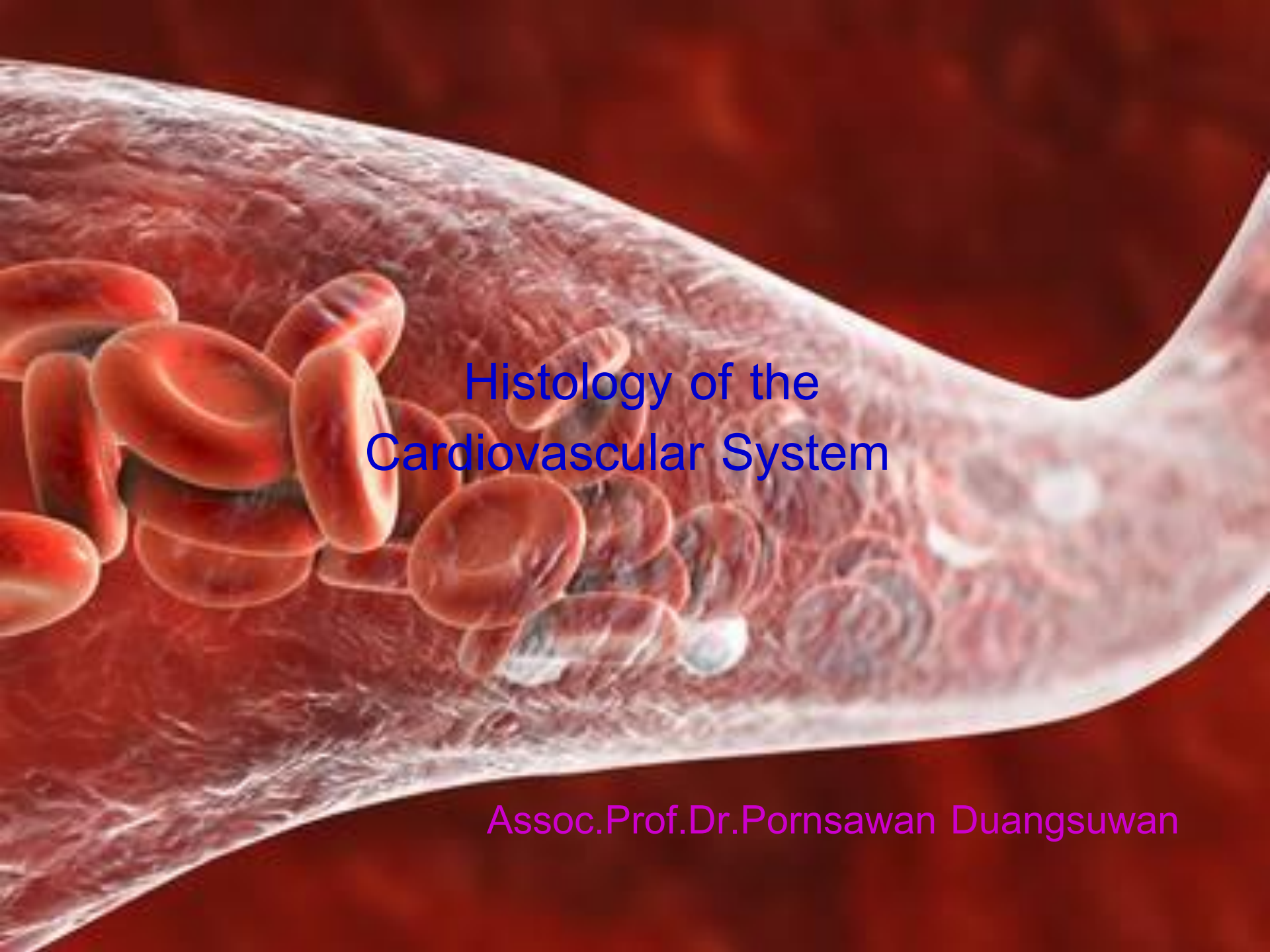


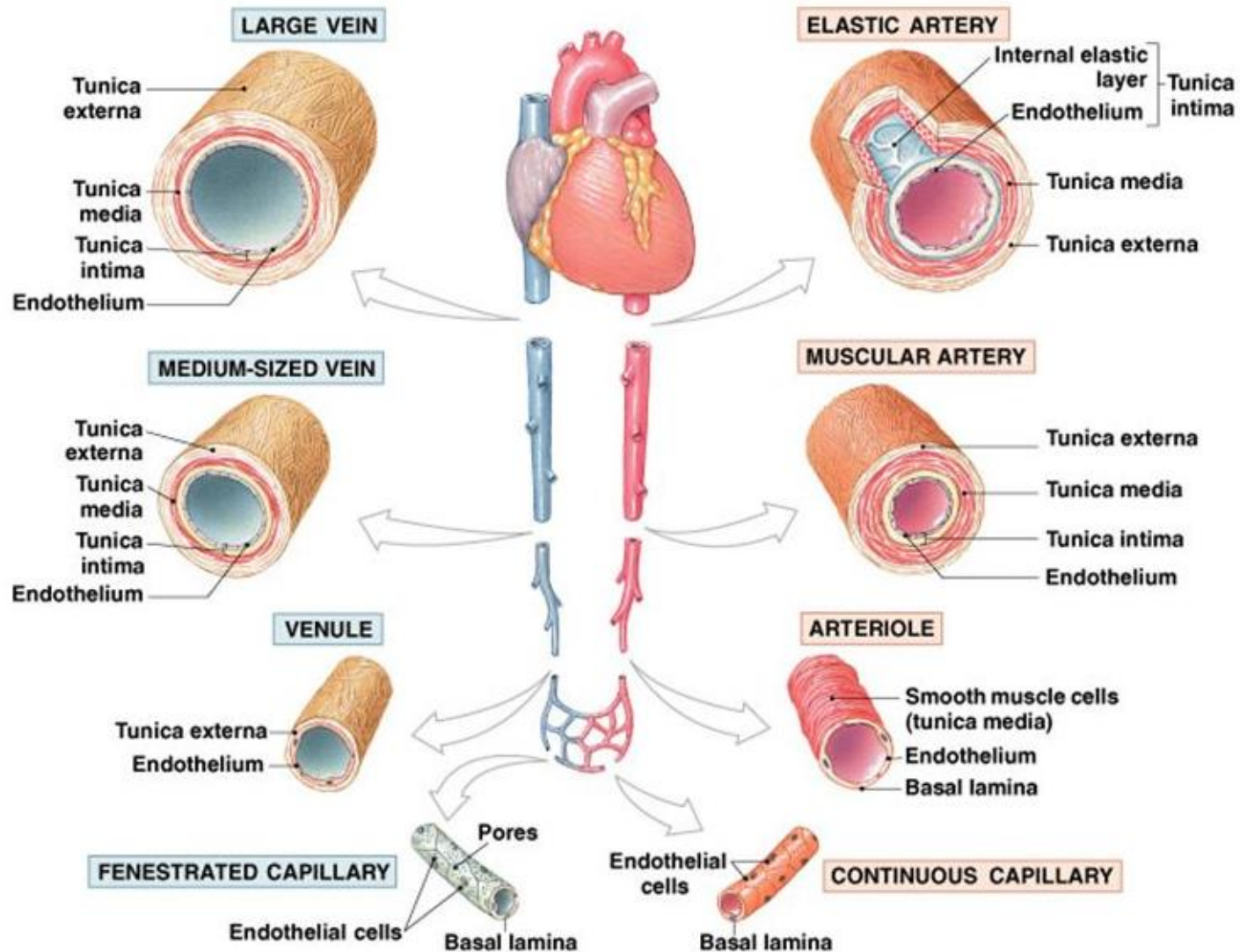


Histology of the Cardiovascular System

Assoc.Prof.Dr.Pornsawan Duangsuwan

Cardiovascular system

Venous system

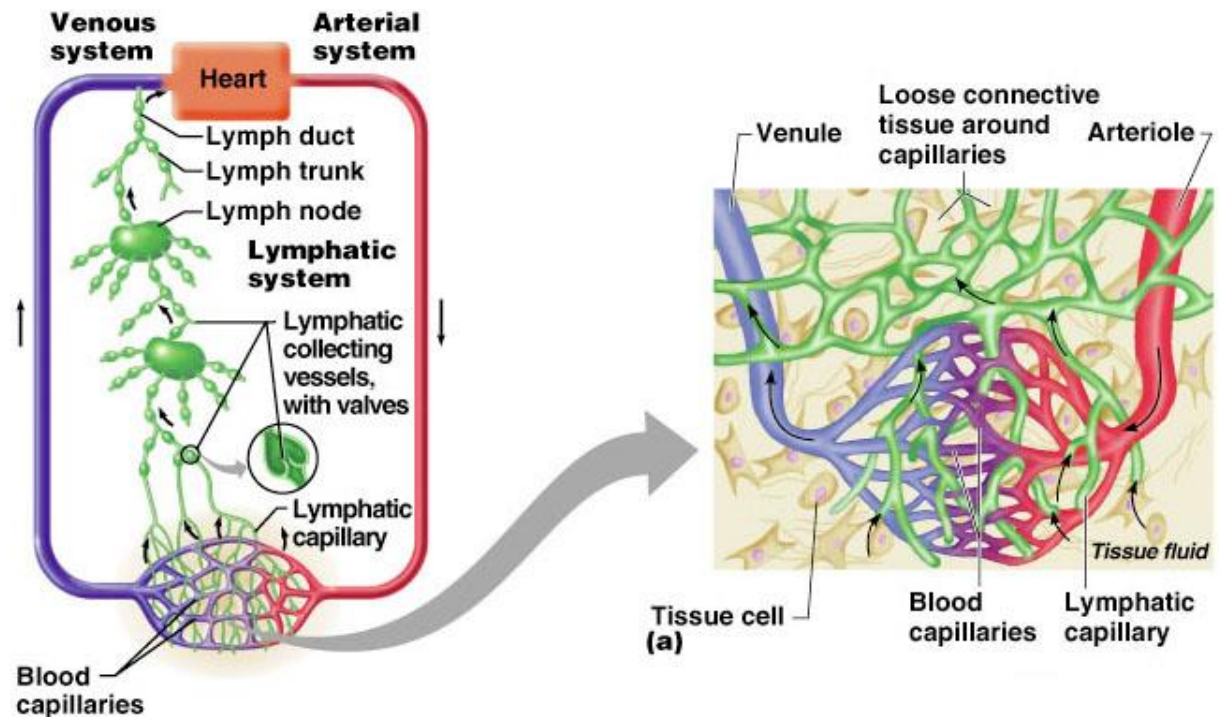


Arterial system

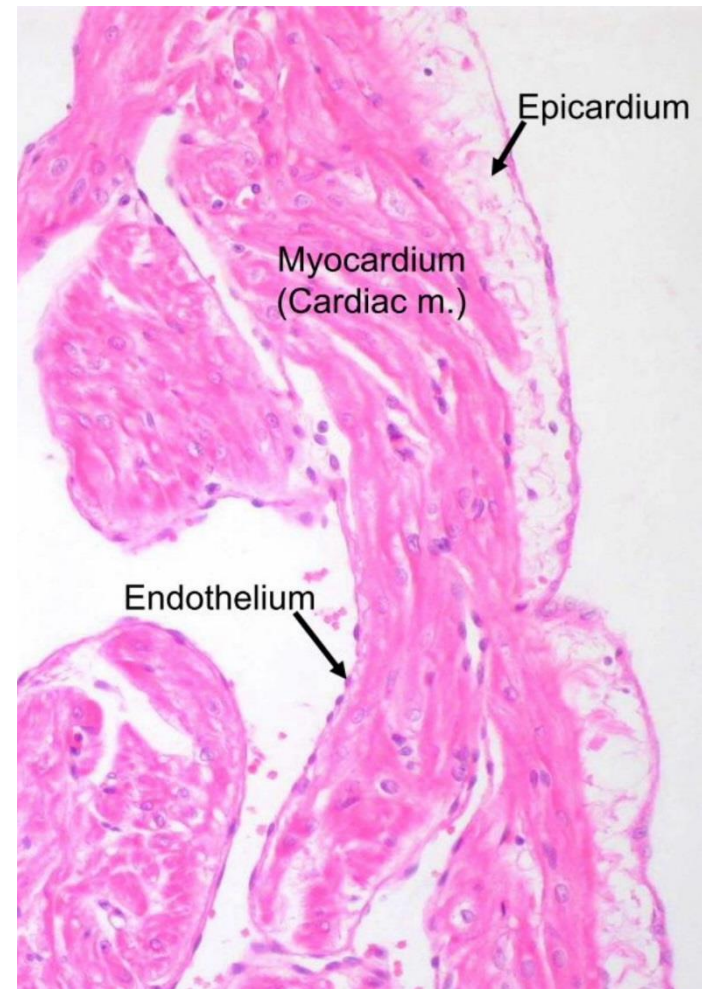
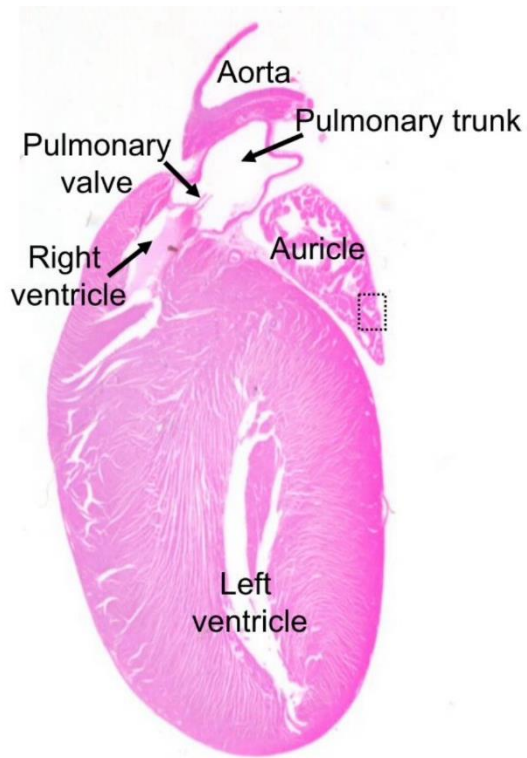
Lymph and Lymphatic Vessels

- **Lymph**: interstitial fluid once it has entered lymphatic vessels
- Returns interstitial fluid and leaked plasma proteins back to the blood

Vein
↑
Lymph node
↑
Lymph vessels
↑
Lymph capillaries
↑
Intercellular fluid

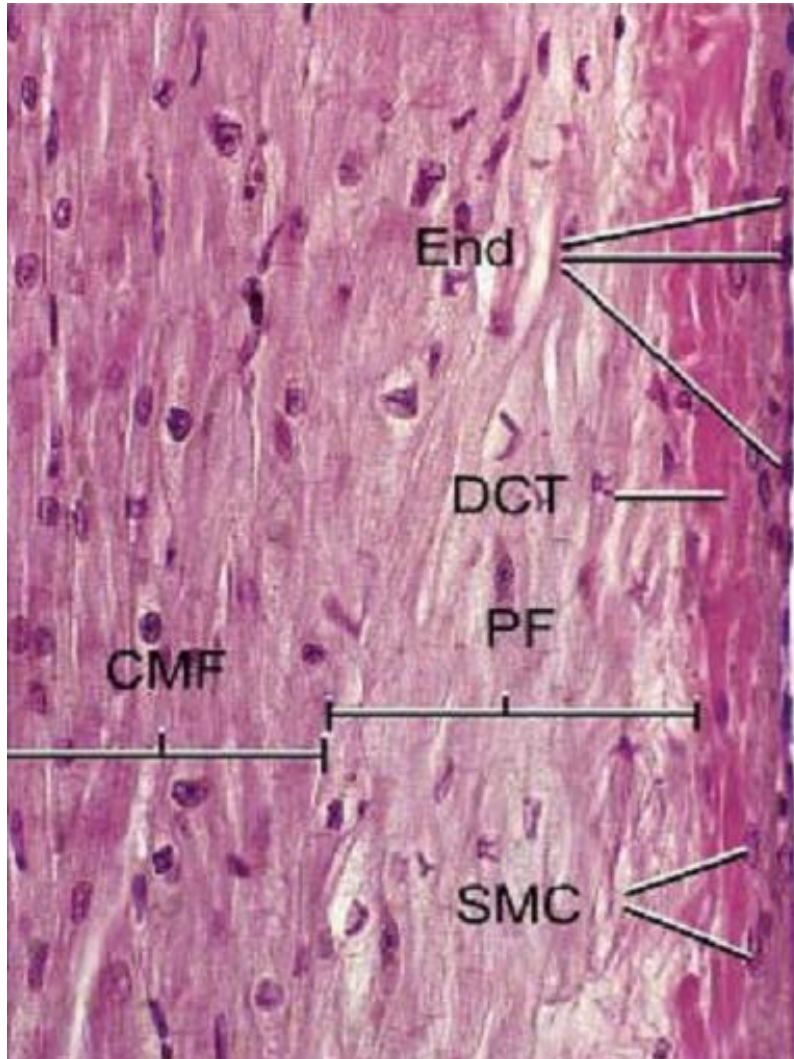


Layers of the heart Wall



1. Endocardium
2. Myocardium: cardiac muscle
3. Epicardium: visceral layer of serous pericardium

Endocardium



1. Endothelial layer

- squamous inner layer

2. Subendothelial layer

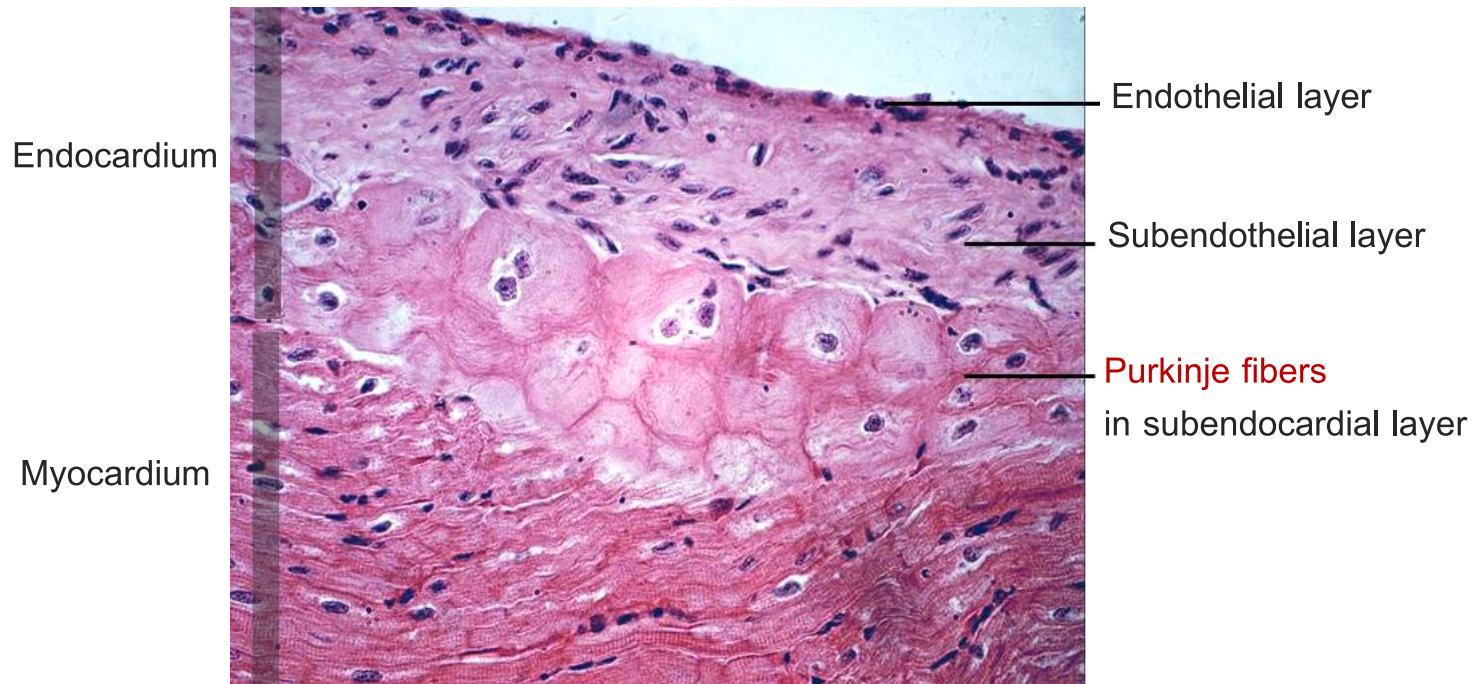
- some dense CNT, smooth m.

3. Subendocardium

- Purkinje fibers

Ventricular wall: Purkinje fibers

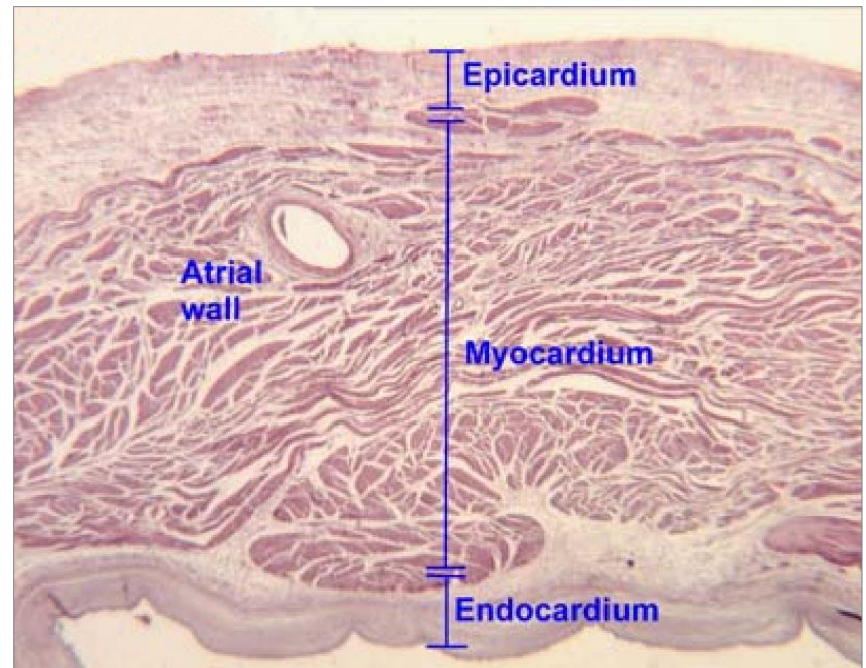
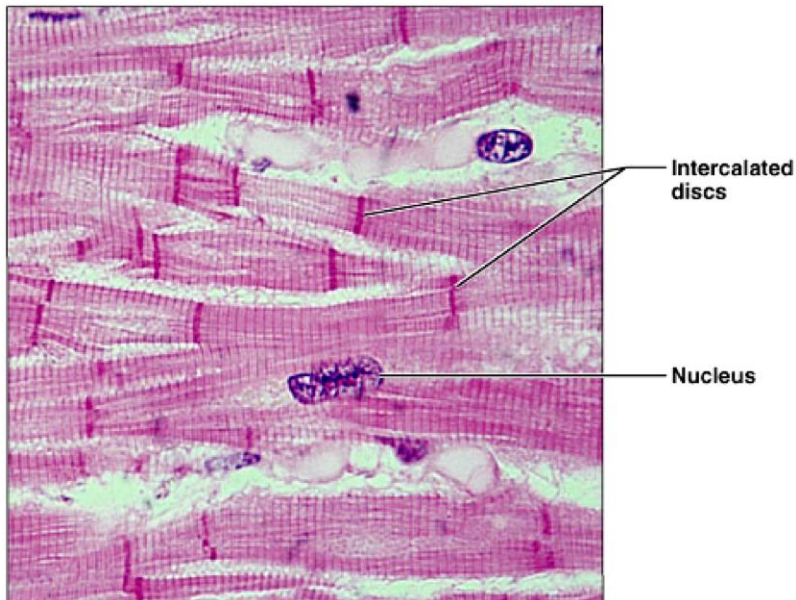
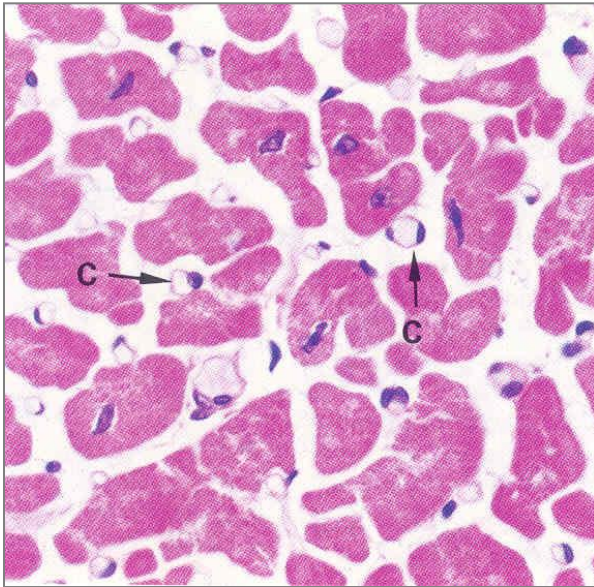
- specialized cardiac muscle cells that generate and conduct electrical signals in heart
- *subendocardium of ventricle*
- larger size, less intense staining
- fewer myofibril, prominent nucleus



Myocardium

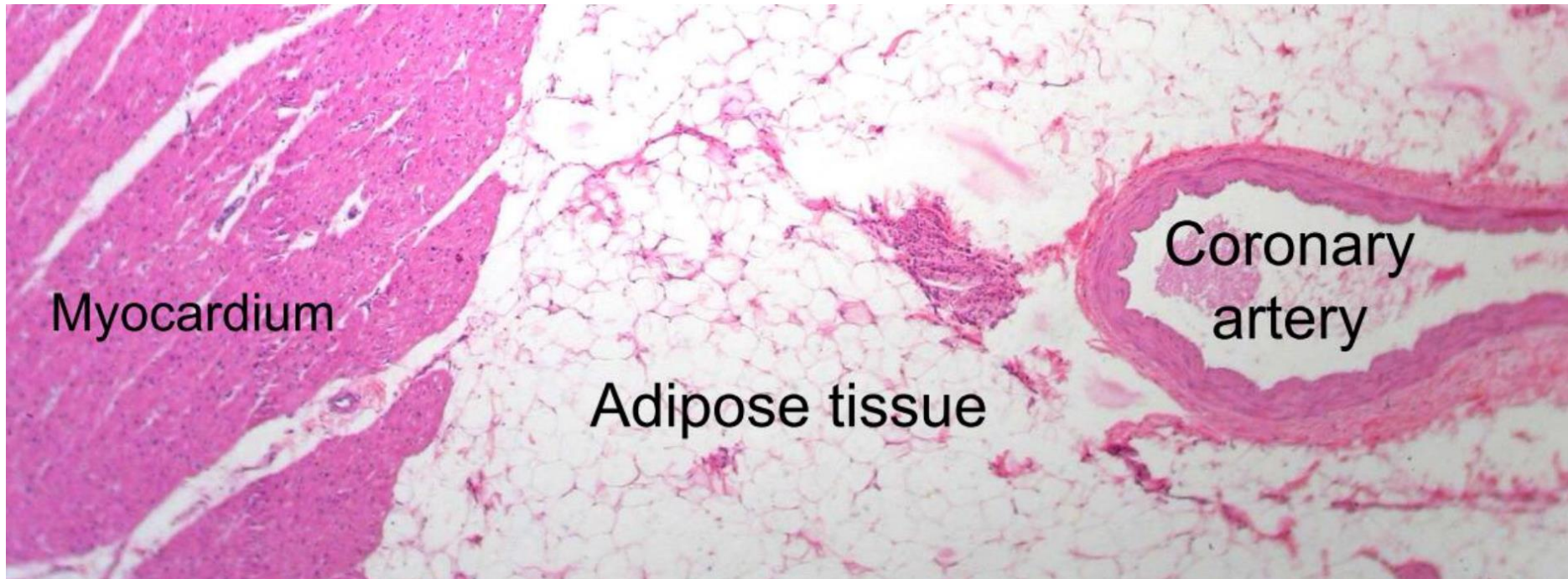
Cardiac muscles

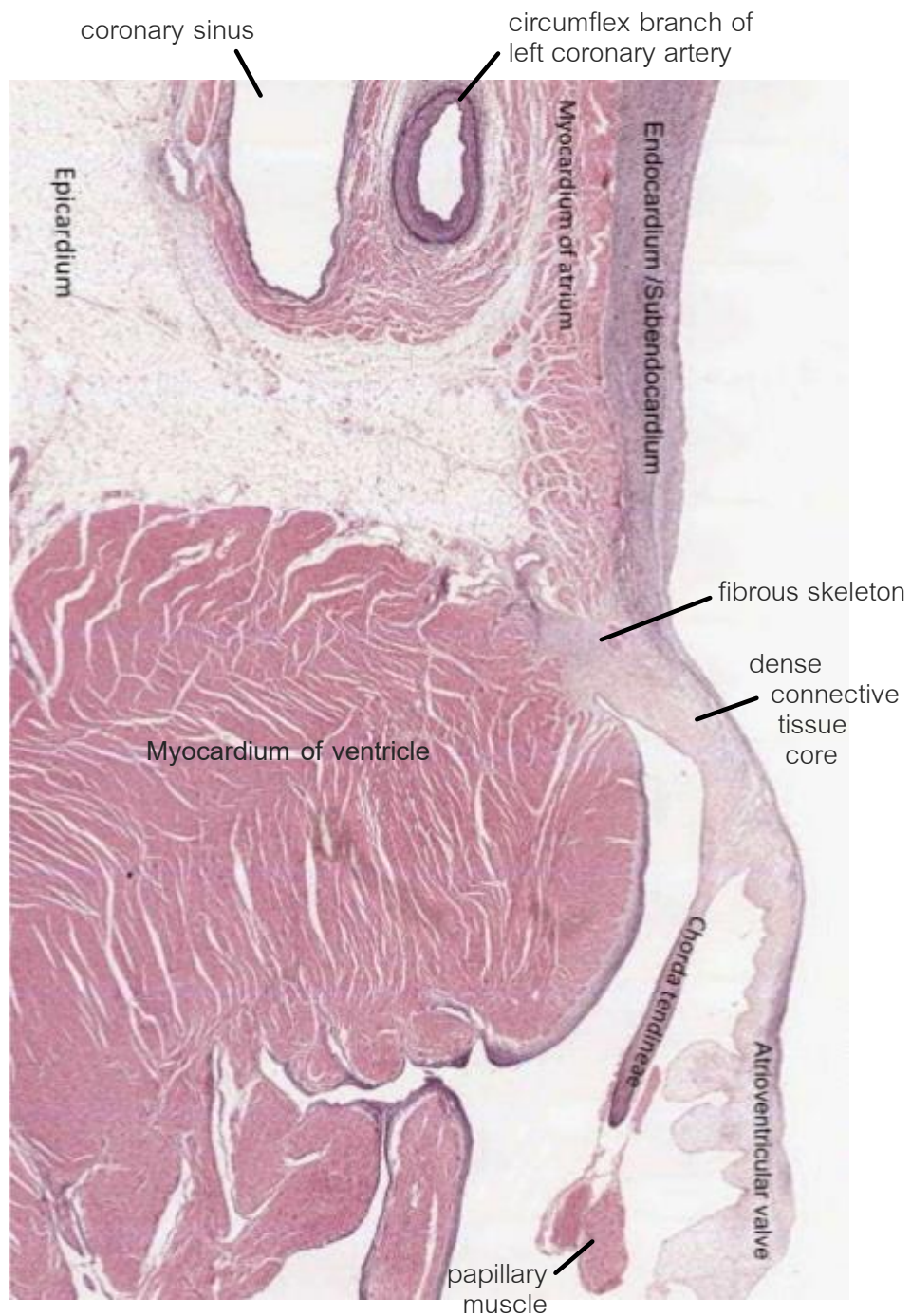
- branching cell
- single central nucleus
- visible striation
- intercalated discs



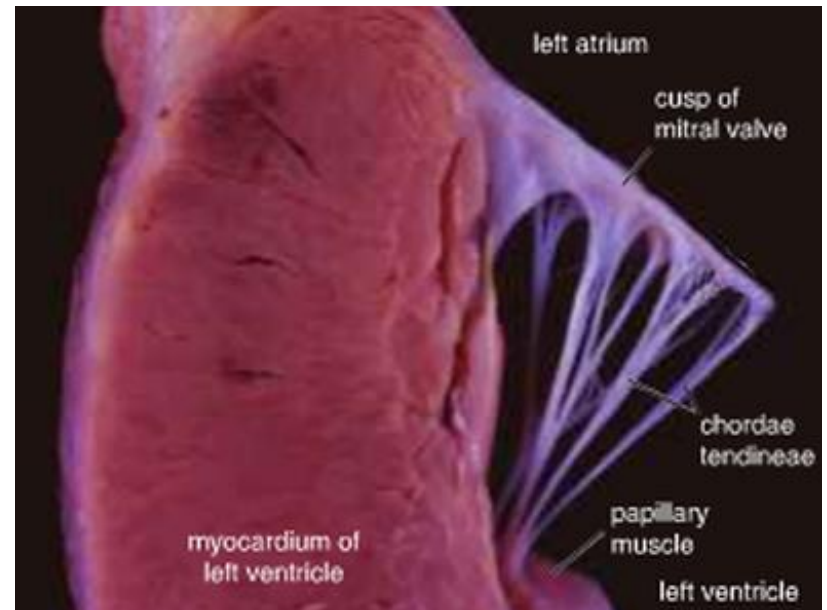
Epicardium

- Visceral layer of serous pericardium
- Layer of mesothelial cells and some connective tissue
- Fat, vessels



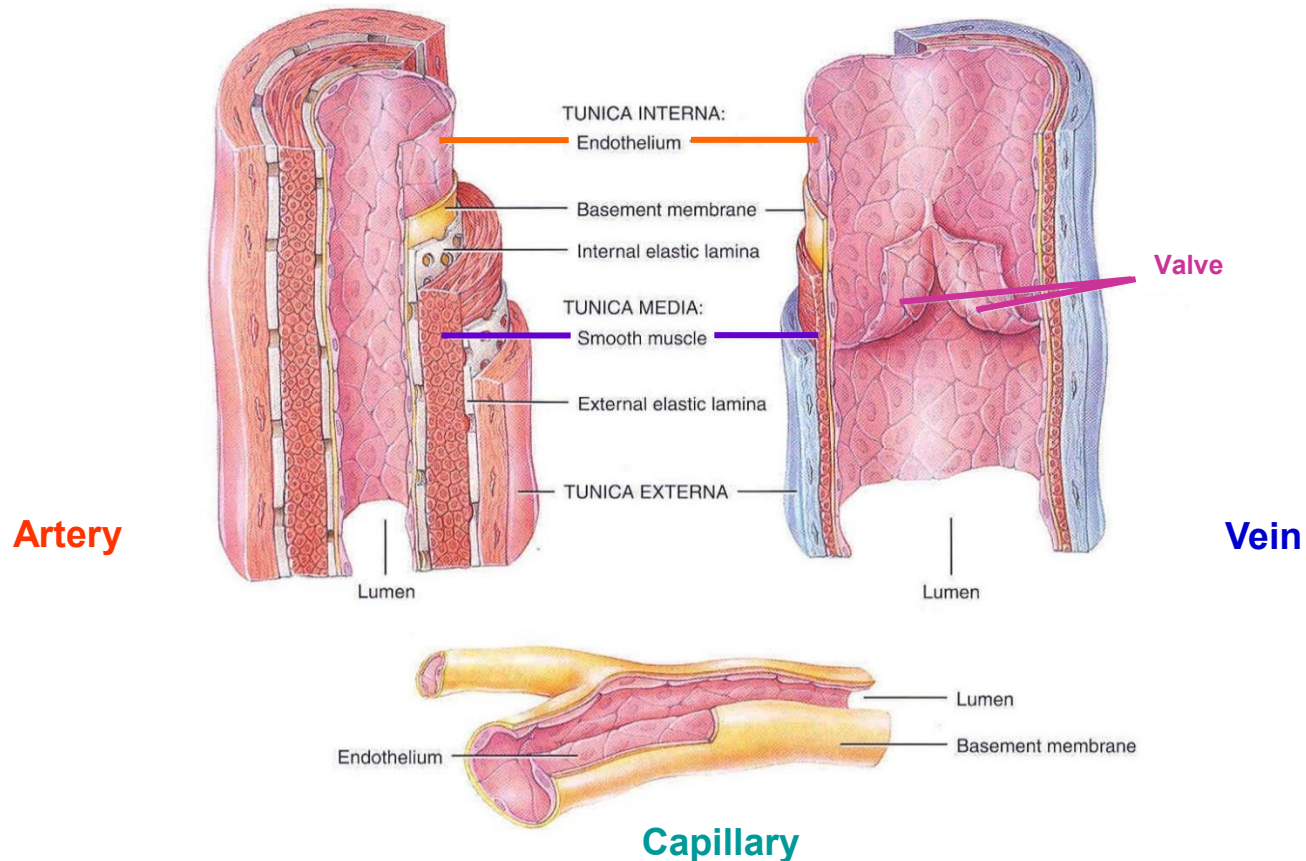


Atrio-ventricular Valves

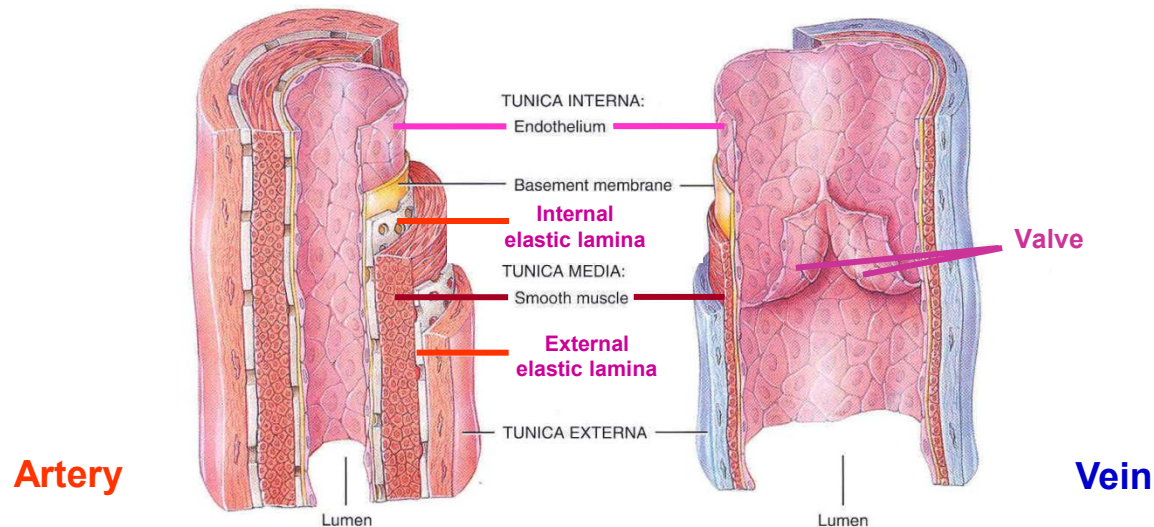


Basic structure of blood vessels

- **Arteries** and **veins** are composed of three layers (tunics):
tunica interna, tunica media, and tunica adventitia
- Capillaries are composed of endothelium with sparse basal lamina



Generalized Structure of Blood Vessels



Tunica intima (interna)

1. Endothelial cell lining
2. Subendothelial layer
3. Internal elastic lamina

Tunica media

1. Smooth muscle cells, collagen fibers, and ground substance
2. Elastin in the form of fenestrated elastic lamellae (esp elastic arteries)
3. External elastic lamina

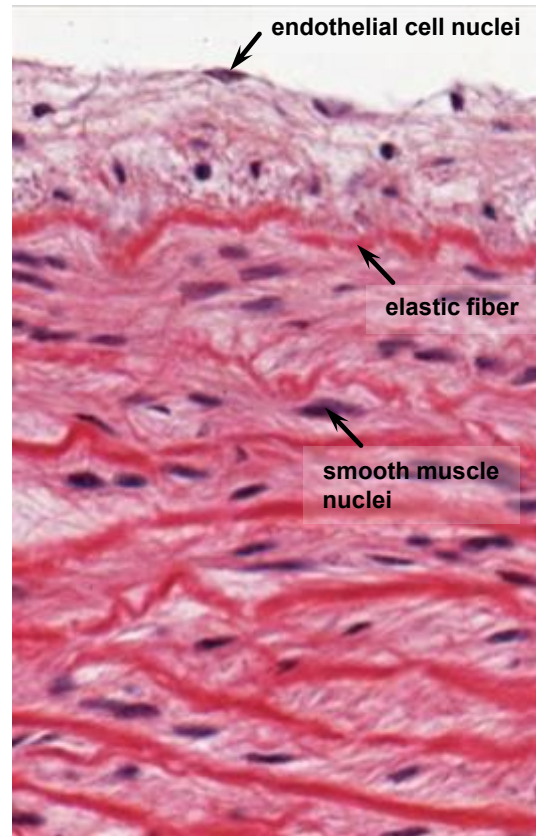
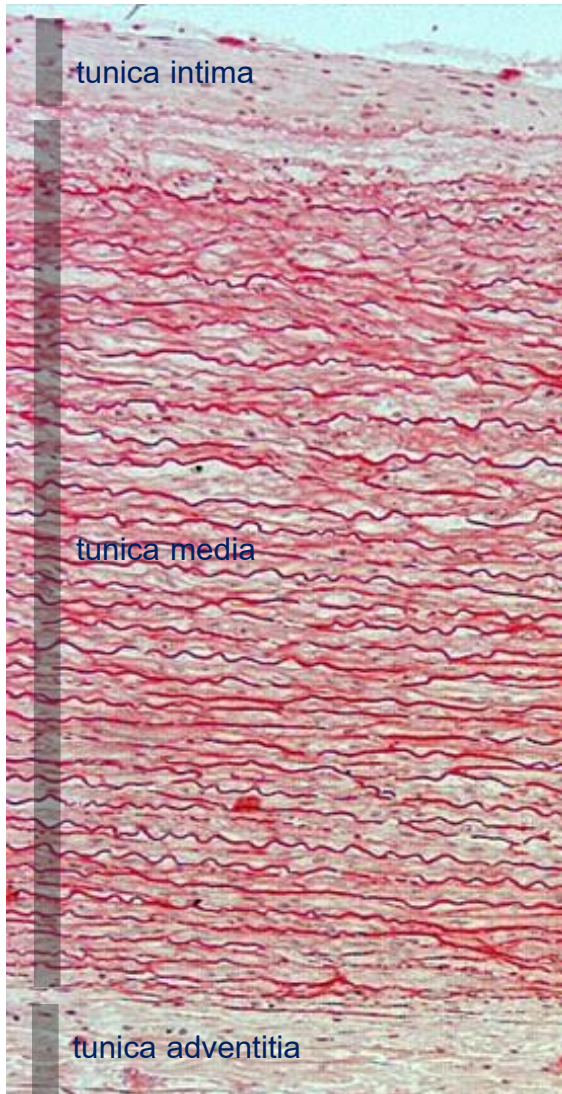
Tunica adventitia

1. Mostly collagenous fibers
2. Elastic fibers (not lamellae)
3. Fibroblasts and macrophages
4. Vasa vasorum

Microscopic Anatomy of Blood Vessels

Large (elastic) artery

- Aorta and its main branches, pulmonary trunk
- Help to continuous blood flow



TI: thin, composed of endothelial cells and underlying supporting tissue

TM: 40-60 *distinct concentric fenestrated elastic laminae*, smooth muscle, collagenous tissue

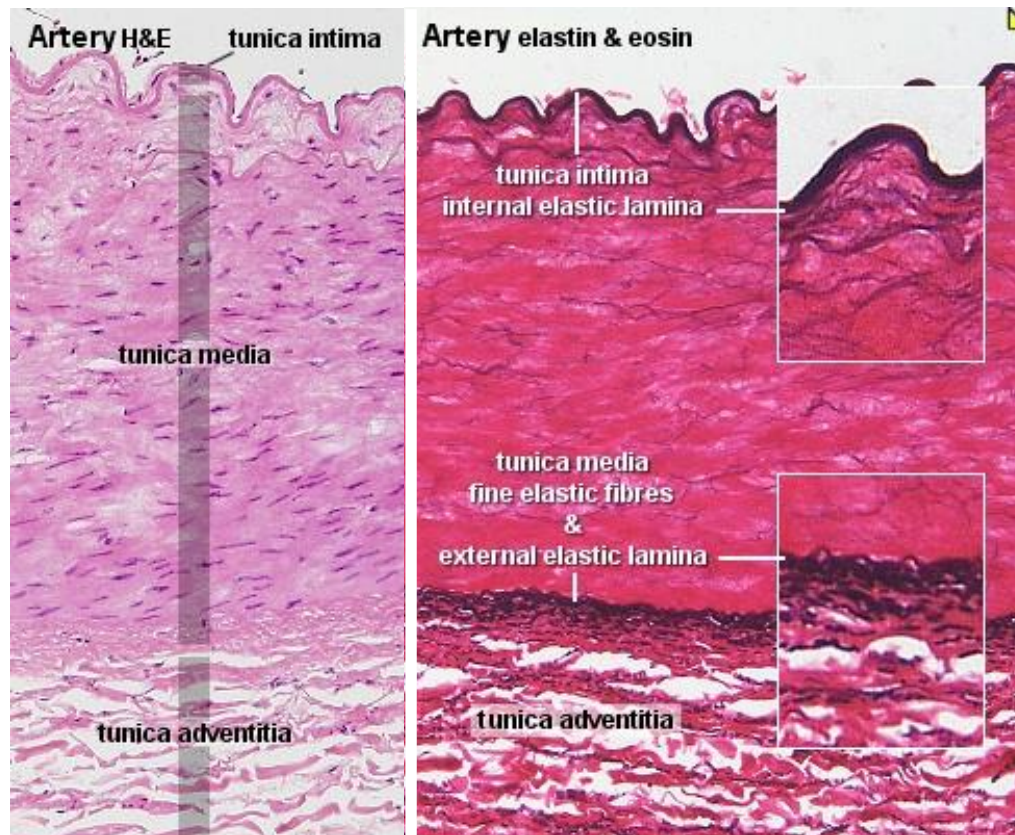
TA: thin, mostly connective tissue, small blood vessel called *vasa vasorum*

unclear internal & external elastic lamina

Medium (muscular, distributing) artery



- External, internal carotid arteries
- Brachial, radial, ulnar, femoral arteries
- Renal, superior-inferior mesenteric arteries
- Intercostal artery



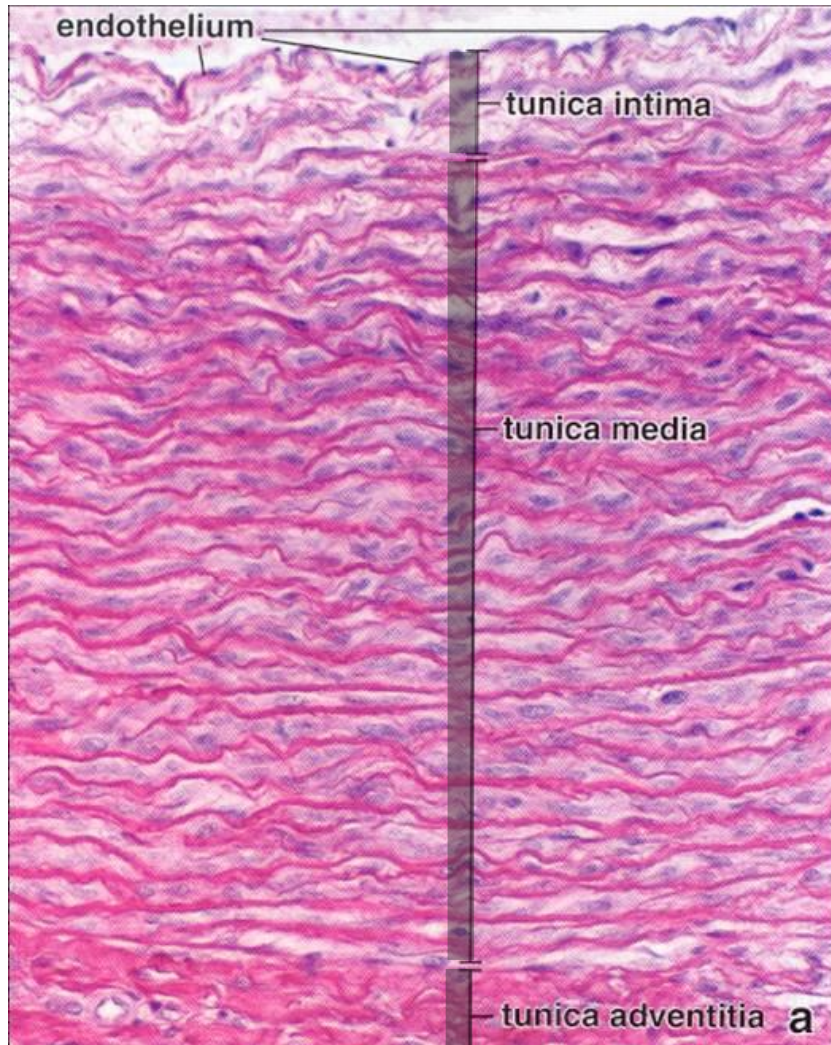
TI: thin subendothelial layer, folded
prominent internal elastic lamina

TM: *25- 40 circular smooth muscle layers*, few elastic & collagen fibers

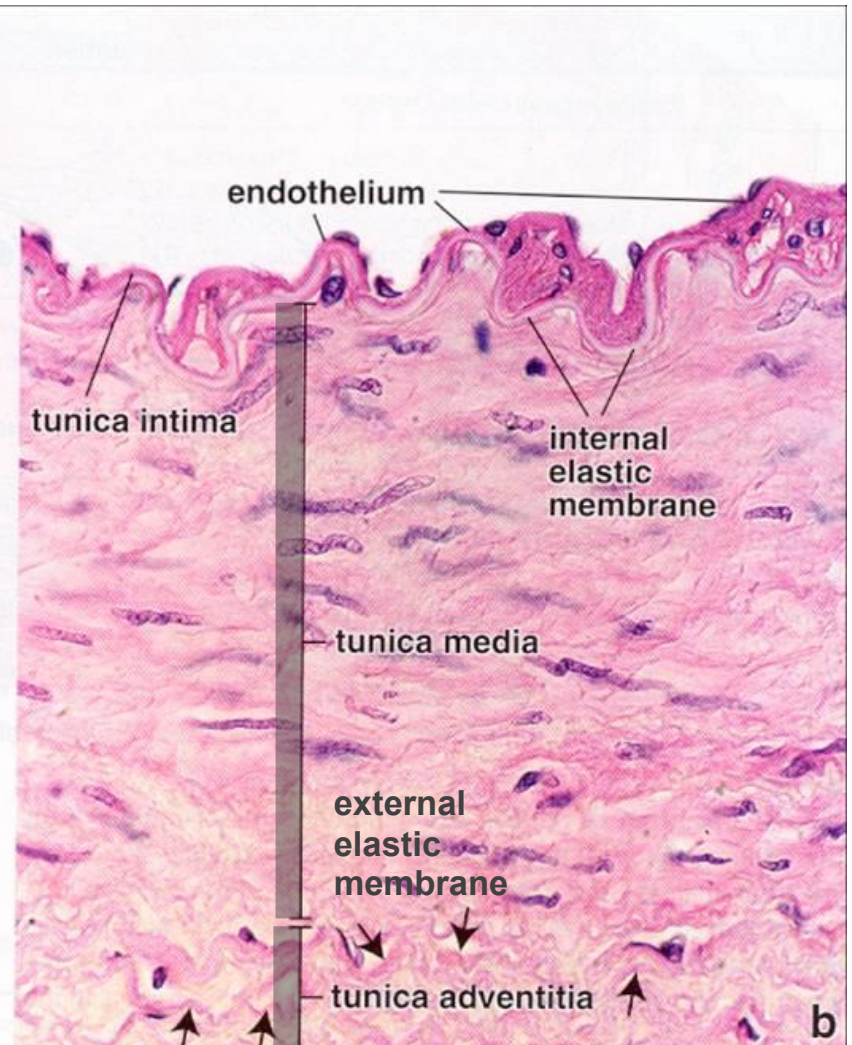
external elastic lamina separates TM from TA (indistinct in smaller muscular arteries)

TA: Thickness may be equal to TM, vasa vasorum

Large (elastic) artery



Medium (muscular) artery



Arteriole



- In tissues and organs
- Contraction of smooth muscle cells constricts the lumen of the arteriole
- Increase vascular resistance and reduce the flow of blood

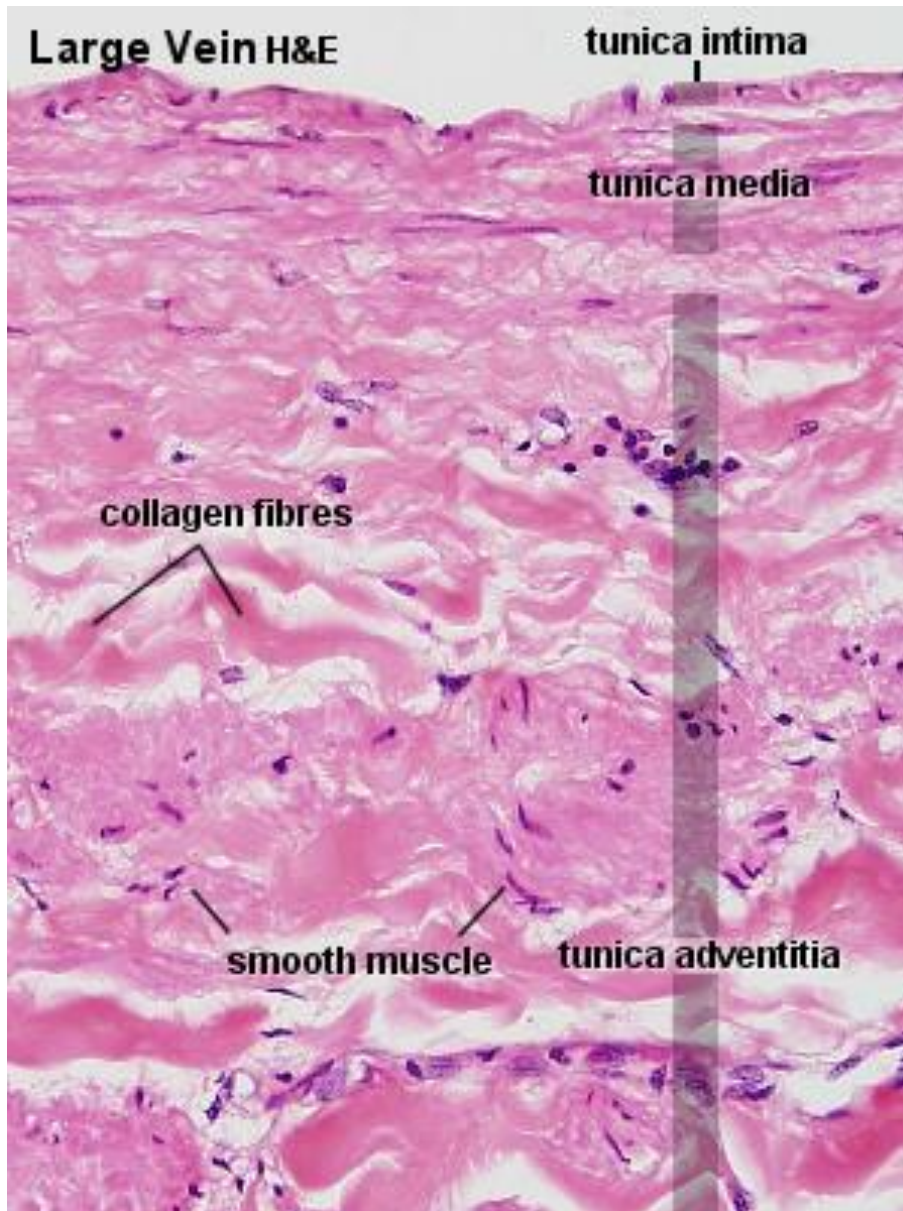
Three layers may be more difficult to define

Wall is relatively thicker than the lumen

TI: only *endothelium*, no subendothelial layer, absent internal & external elastic lamina

TM: *only 1-5 layers* of smooth muscle cells, some elastic fibers

TA: thin, longitudinally arranged collagen and elastic fibers



Large vein

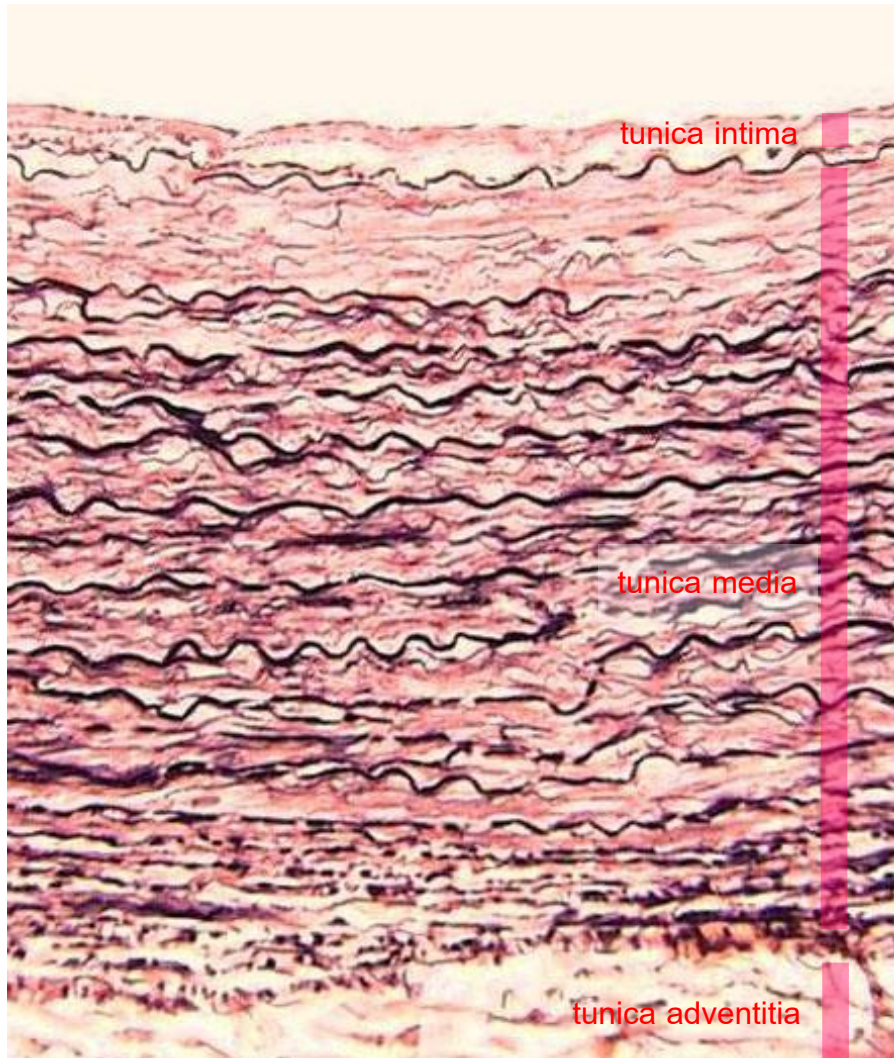
- SVC, IVC
- Internal jugular vein
- Portal vein

TI: well developed, endothelium and subendothelial connective tissue

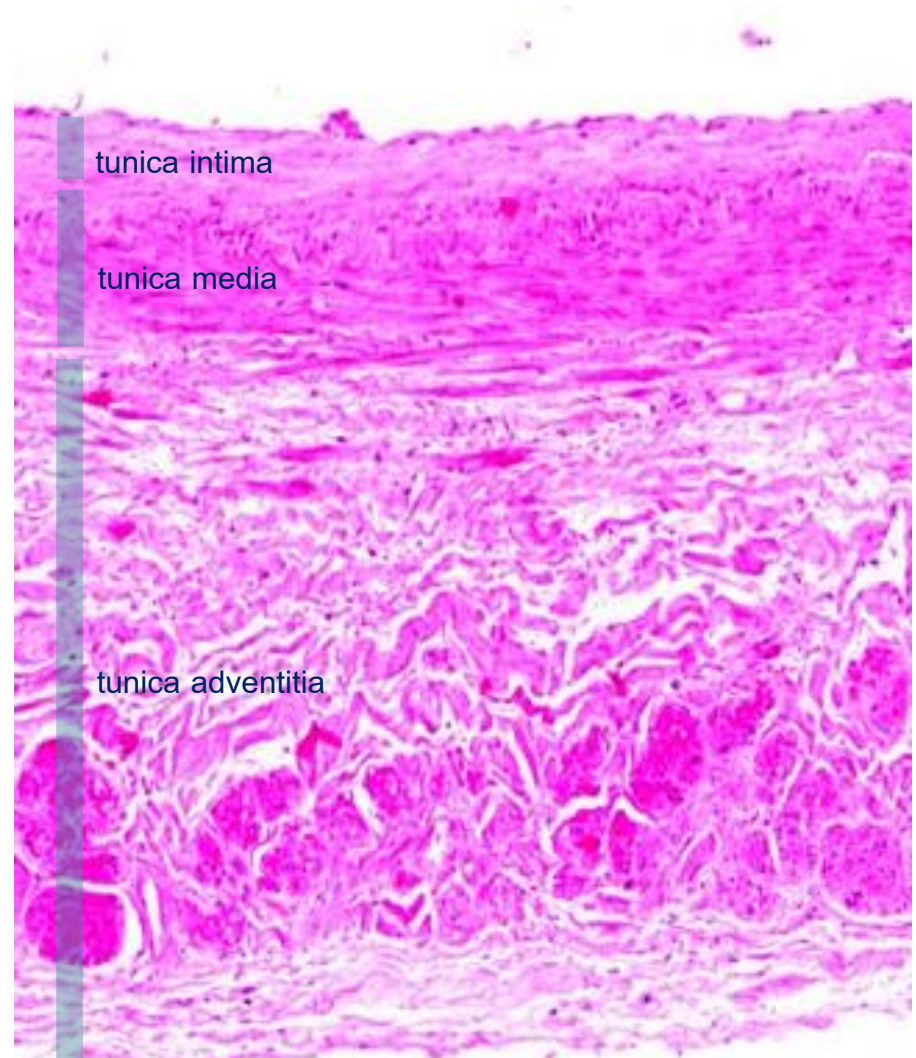
TM: *relatively thin*, compared to the thick tunica adventitia

TA: well developed adventitia, *longitudinally-oriented bundles of smooth muscle* facilitate shortening and elongation of the vena cava with respiration

Large (elastic) artery

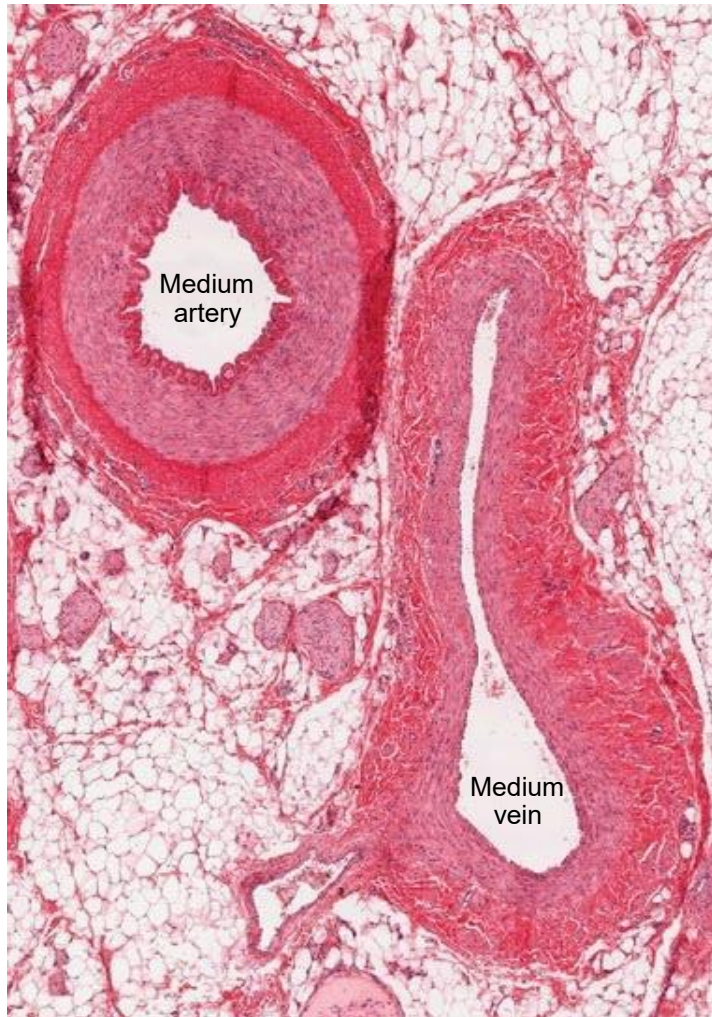


Large vein



Medium vein

- Brachial, radial, ulnar, femoral veins
- Renal, superior-inferior mesenteric veins
- Intercostal vein
- Basilic, cephalic veins



Collapsed larger lumen

Ratio of wall thickness to lumen diameter is less

TI: endothelium and thin layer of subendothelium

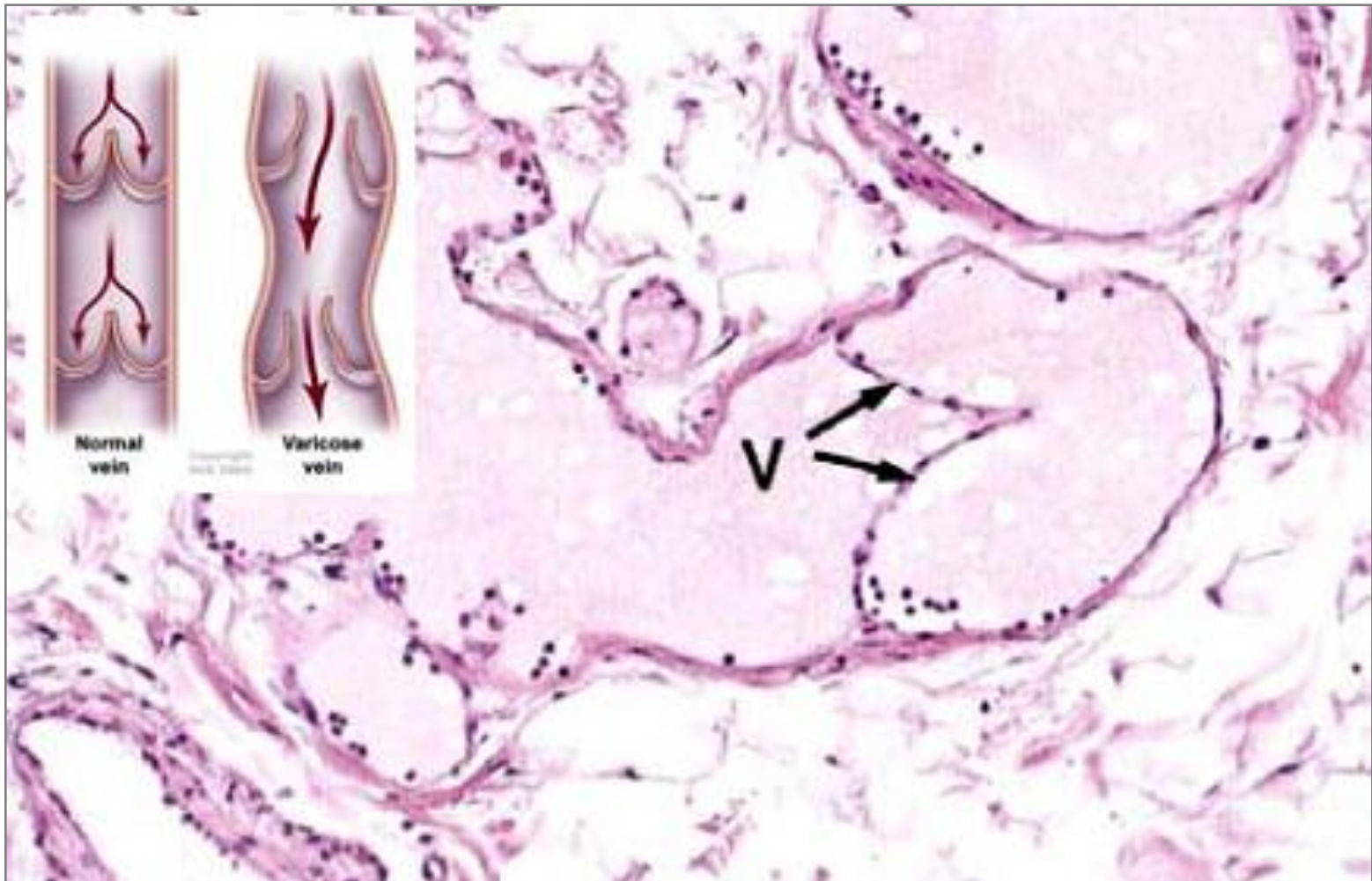
TM: *thin and poorly developed*, few circular smooth muscle fibers layers

predominantly collagen fibers, less elastic fibers

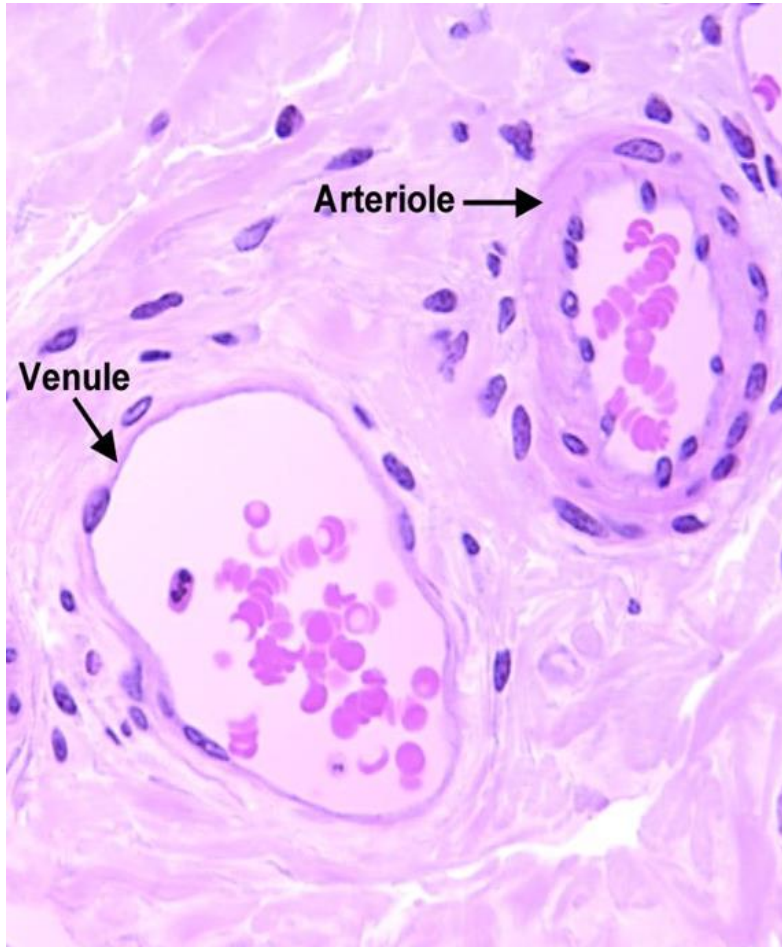
TA: loose fibroelastic connective tissue with vasa vasorum

Valve of vein

: inflodings of the tunica intima



Venule



Collapsed larger lumen

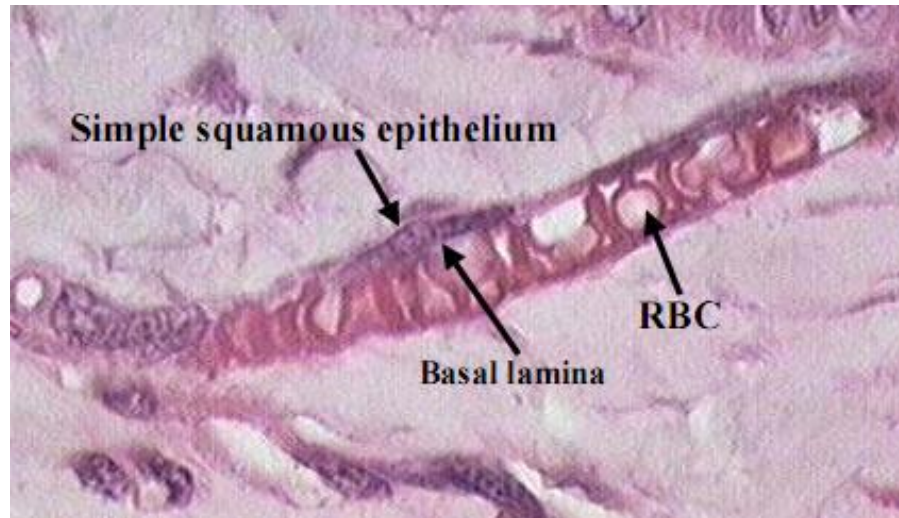
Ratio of wall thickness to lumen diameter is less

TI: single layer of flattened endothelial cells

TM: sparse connective tissue

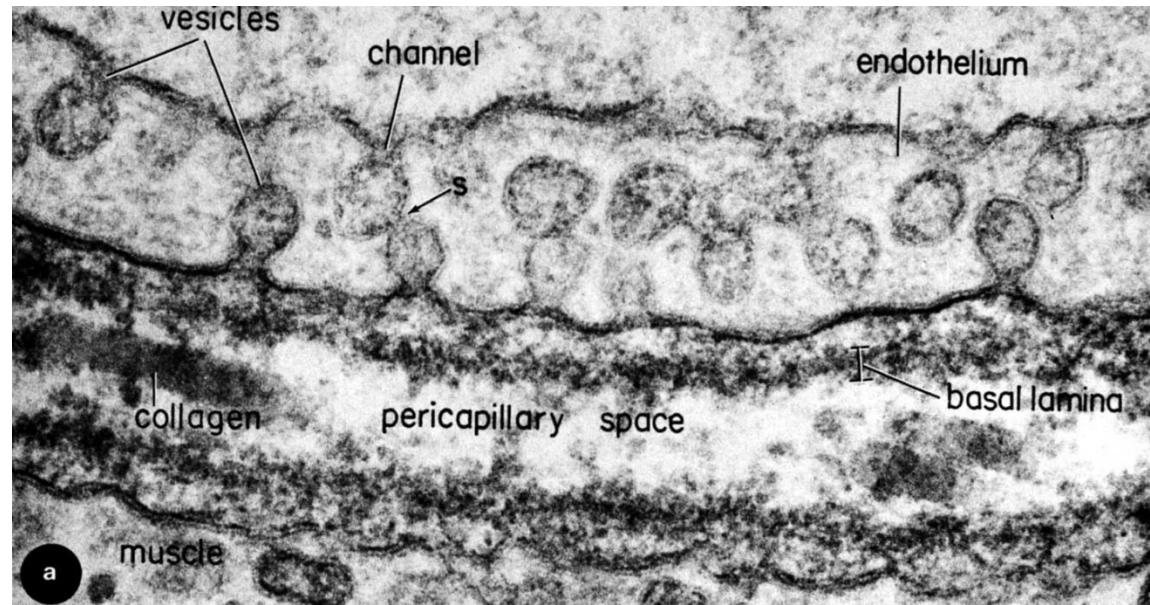
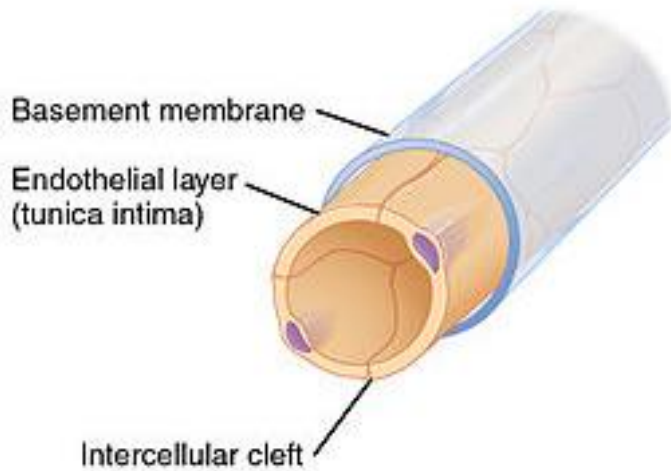
TA: some collagen and a few fibroblast

Capillaries



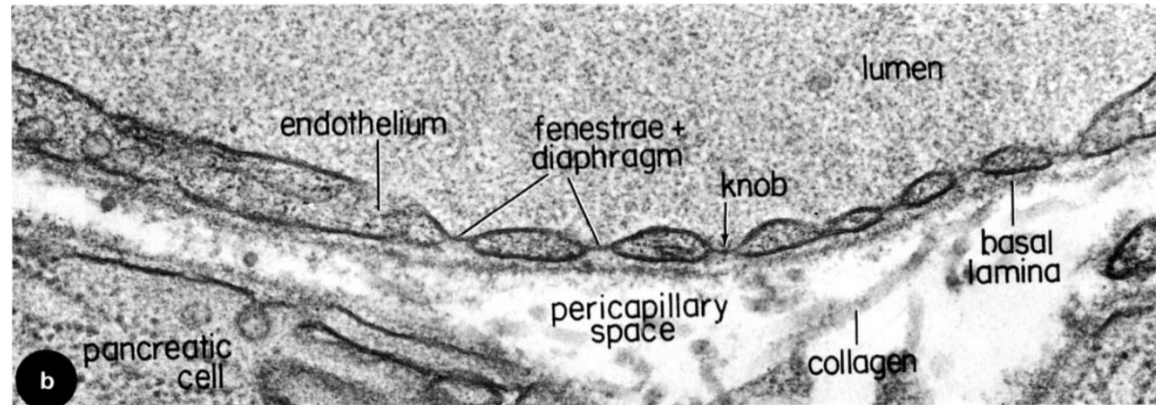
- Capillaries are the smallest blood vessels
 - Walls consisting of a thin tunica interna, one cell thick
 - Allow only a single RBC to pass at a time
 - Pericytes on the outer surface stabilize their walls
- There are three structural types of capillaries:
continuous, fenestrated, and sinusoids

Continuous capillary



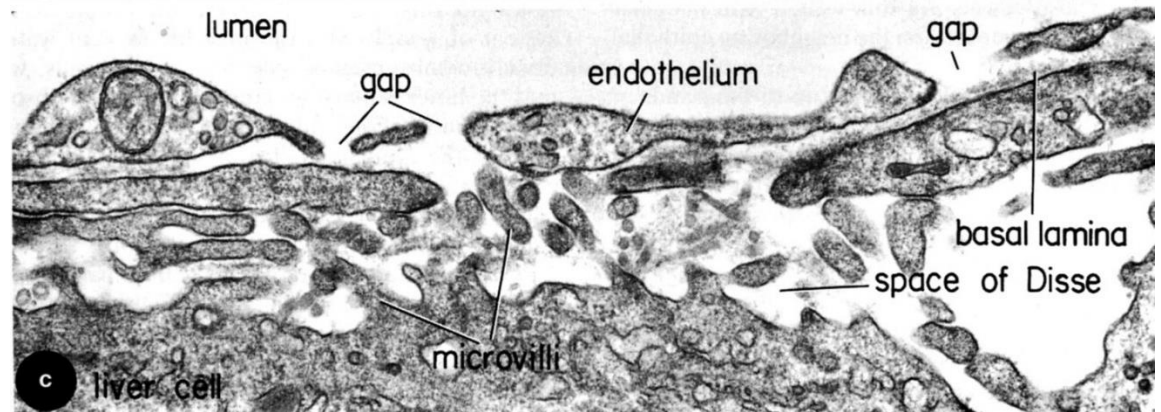
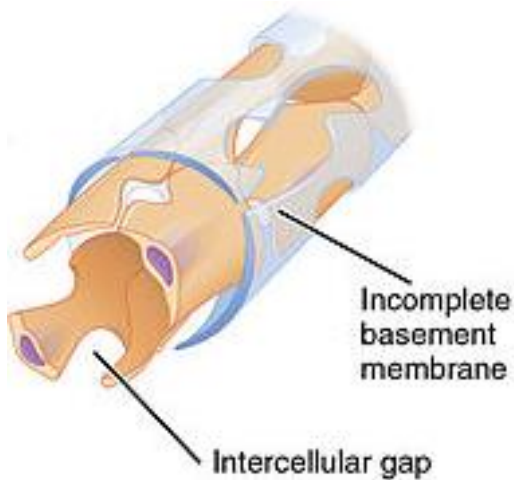
- Endothelial cells that provide an uninterrupted lining
- Adjacent cells are held together with tight junctions
- Intercellular clefts of unjoined membranes that allow the passage of fluids
- muscles, connective tissue, lung, skin

Fenestrated capillary



- Found wherever active capillary absorption or filtrate formation occurs (e.g., small intestines, endocrine glands, and kidneys)
- Characterized by:
 - Gaps between endothelial cells (fenestrations),
 - Basement membrane is continuous
 - Greater permeability to solutes and fluids than other capillaries

Sinusoidal capillary

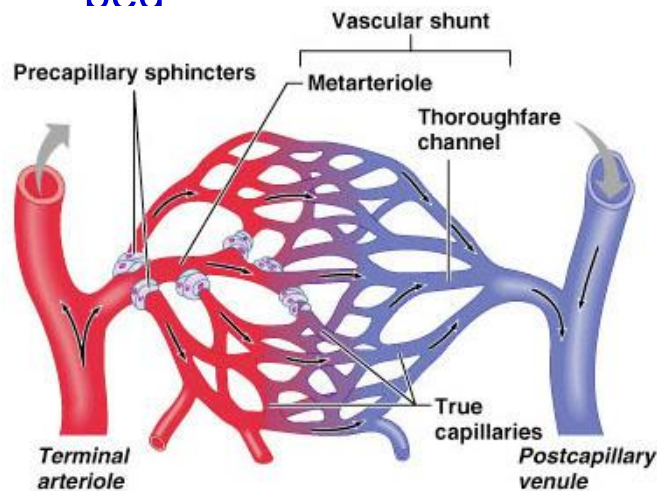


- Highly modified, leaky, fenestrated capillaries with large lumens
- Found in the liver, bone marrow, lymphoid tissue
- Allow large molecules (proteins and blood cells) to pass between the blood and surrounding tissues
- Blood flows sluggishly, allowing for modification in various ways

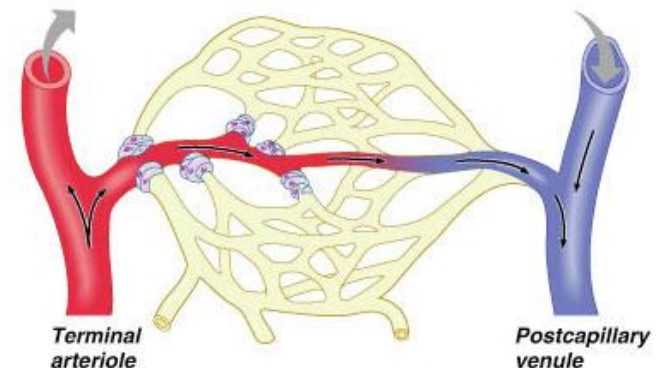
Capillary Beds

A microcirculation of interwoven networks of capillaries, consisting of:

- Vascular shunts: metarteriole–thoroughfare channel connecting an arteriole directly with a postcapillary venule
- True capillaries: 10 to 100 per capillary bed, capillaries branch off the metarteriole and return to the thoroughfare channel at the distal end of the bed



1. Sphincters relaxed: blood flowing through capillary bed



2. Sphincters contracted: blood flowing through thoroughfare channel

Conclusions

Elastic a.

1. TM พบ elastic fiber มาก
2. IEL, EEL ไม่ชัด

Medium (muscular) a.

1. แบ่ง 3 ชั้นชัดเจน
2. IEL หยักชัดที่สุด
3. TM พบ smooth m. จำนวนมาก

Arteriole

1. แบ่ง 3 ชั้นไม่ชัด
2. endothelial cell 3-5 cells
3. smooth m. ล้อมรอบ ไม่เกิน 5 ชั้น

Vein

1. รูปร่างไม่แน่นอน
2. ผนังบางเมื่อเทียบกับ lumen
3. แบ่ง 3 ชั้นไม่ค่อยชัดเจน

Large v.

1. TI, TM เจริญไม่ค่อยดี
2. TA พบ smooth m. จำนวนมาก

Medium v.

1. ผนังบางเมื่อเทียบกับ lumen
2. TM พบ smooth m. และ CNT
3. TA หนาชัด

Venule

1. endothelial cell บางๆ เรียงเป็นระยะบน CNT

THE END

